

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457388.7267 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.114 +/- 0.042
Transit depth (Rp/Rs)^2	1.29 +/- 0.96 [%]
Orbital Inclination (inc)	87.6 +/- 2.89
Airmass coefficient 1 (a1)	1.82 +/- 1.01
Airmass coefficient 2 (a2)	-0.34 +/- 0.29
Scatter in the residuals of the lightcurve fit is	54.97 %
Best Comparison Star	None
Optimal Aperture	1.93
Optimal Annulus	5.75
Transit Duration (day)	0.104 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.114 ± 0.042 vs 0.116 (deviation 0.05σ)
Duration fit vs published	0.104 d vs 0.09 d (deviation 0.58σ)
Mid-time O–C	3.9 min
Depth SNR	2.7
Residual scatter	54.97 %

Light curve

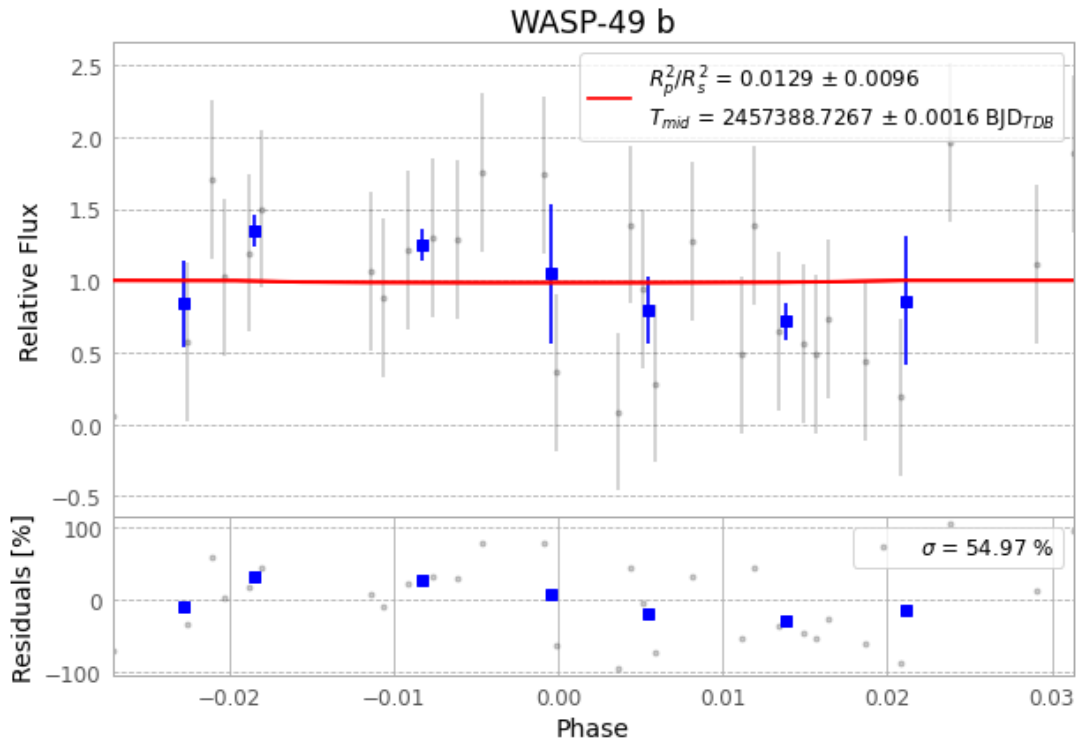


Figure 1 — FinalLightCurve_WASP-49 b_2015-12-31.png (SHA-256 28ea447ea26cbd04...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [400, 74], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b5a60f1b865654ec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), WASP-49160101032412.FITS → WASP-49160101072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-160101.log (c2dd68643a5f...), FinalLightCurve_WASP-49 b_2015-12-31.png (28ea447ea26c...), FinalLightCurve_WASP-49 b_2015-12-31.csv (592034445ca0...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 15:26Z.